

Name: _____

Materials that you may use during the exam:

1. One page (double sides) cheat sheet (A4). ONLY equations/formulas are allowed.
2. NOT open book
3. NOT open notes (including the printed handouts the instructors distributed in class)
4. NOT open to any other resources (including homework, quizzes, practical exercises).

** Violation of the above policy will result in significant grade deduction.*

A (15 pts) Multiple choice (3 pts each).

1. The percentage humidity is less than the percentage relative humidity only atpercent humidity.
(a) 0
(b) 100
(c) both a & b
(d) none of these.
2. The falling rate period in a drying of a solid is characterized by
(a) increase in rate of drying
(b) increasing temperatures both on the surface and within the solid
(c) decreasing temperature
(d) none of these.
3. The temperature of cooled water in a cooling tower is
(a) always less than the wet-bulb temperature of entering air
(b) always greater than the wet-bulb temperature of entering air
(c) always equal to the wet-bulb temperature of entering air
(d) always equal to the dry-bulb temperature of entering air
4. Which of the following is an expression for J_A^* ? _____
 ρ_A is density, v_A is the average velocity of the species.
(a) $\rho_A v_A$
(b) $C_A v_A$
(c) $-CD_{AB}\nabla y_A$
(d) $-\rho_A D_{AB}\nabla \omega_A$
5. What is the maximum possible humidity (kg/kg dry air) of air at 52 °C and at 1 atm pressure?
(a) 0.05
(b) 0.0207
(c) 0.098
(d) 0.042.

B (15 pts) Short answer (3 pts each).

1. How concentration of the component can be obtained from the total pressure?

2. What are wet bulb temperature and adiabatic saturation temperature?

3. What is the difference between drying and evaporation?

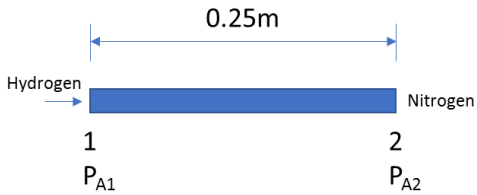
4. How separation of components takes place in membrane?

5. How mass transfer coefficient related to film thickness of liquid?

C. (20 pts) Nitrogen gas stored in a 30 m^3 container at 340 K and 101.3 kPa is saturated with water vapor. Partial pressure of water vapor is 27.35 kPa . Determine the following of the gas mixture: a) Mole fraction of water vapor. b) Average molecular weight of the mixture. c) Total mass contained in the tank. d) Mass of water vapor in the tank. hint: the molecular mass of nitrogen is 28 and the molecular mass of water is 18.

D. (20 pts) Hydrogen gas (A) is diffusing through a uniform tube 0.25 m long containing nitrogen gas (B) at 1.0132×10^5 Pa pressure and 298 K. At point 1, $P_{A1} = 1.0132 \times 10^4$ Pa and at point 2, $P_{A2} = 0.67 \times 10^4$ Pa. The diameter of the tube is 0.05m. The diffusivity D_{AB} is $0.784 \times 10^{-4} \text{ m}^2/\text{s}$.

Calculate (a) the flux J_A^* at steady state and (b) the mass flow rate of the hydrogen.



E. (30 pts) Calculate (a) the overall mass transfer coefficient, (b) the flux and (c) the rate of removal of the impurity (compound A) at steady state in g/h from the protein solution in a regenerated cellulose membrane dialyzer at 25°C. The membrane is 0.03 mm thick and has an area of 1.5 m². The mass-transfer coefficient on the protein side is $K_{c1} = 1.5 \times 10^{-5}$ m/s and that on the aqueous side is $K_{c2} = 2.8 \times 10^{-5}$ m/s. The permeance of the membrane is 8.42×10^{-6} m/s. The concentration of the impurity in the blood is 0.1g compound A/liter and the that in the dialyzing fluid is very low (assumed as 0).

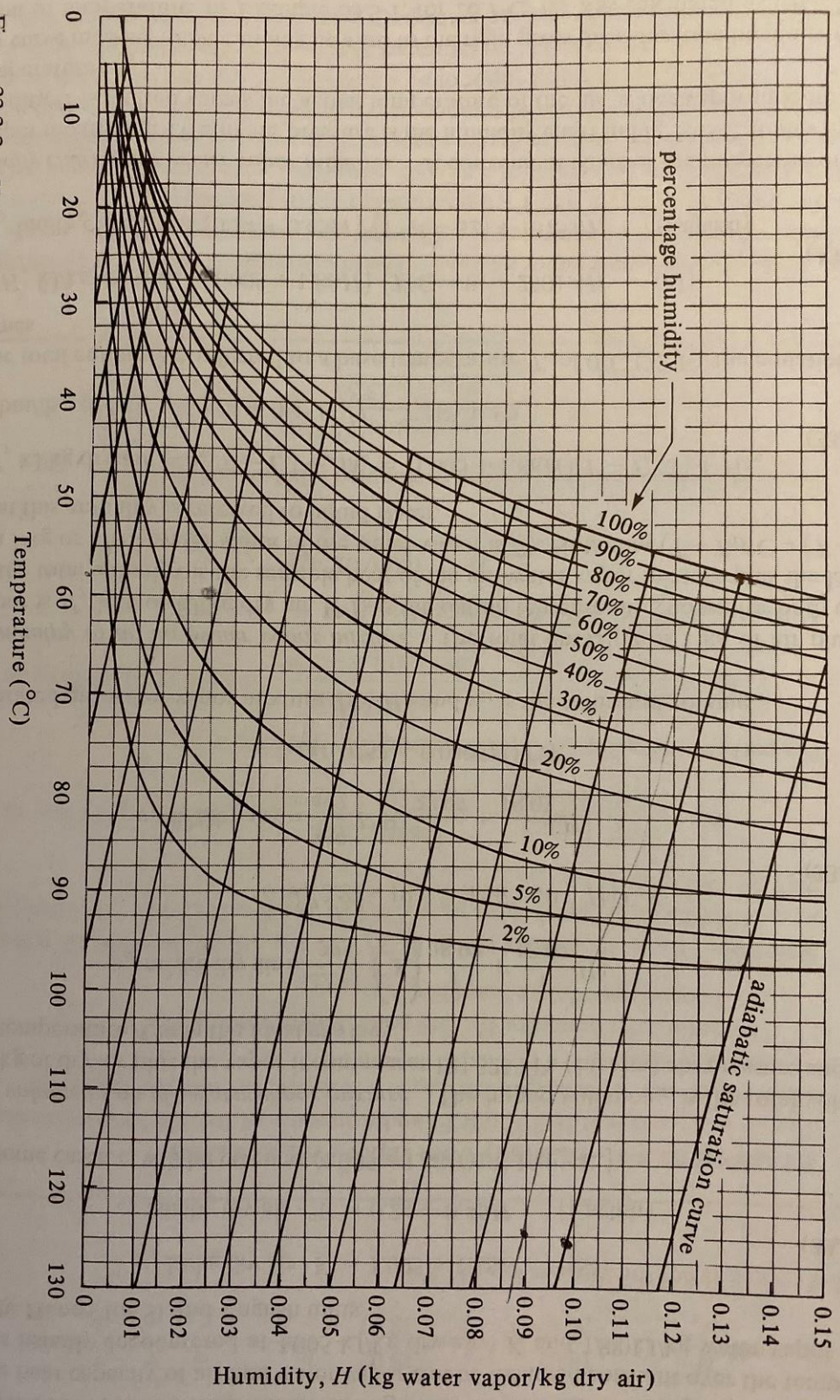


Figure 33.3-2. Humidity chart for mixtures of air and water vapor at a total pressure of 101.325 kPa (760 mmHg). From R. E. Treybal, Mass-Transfer Operations, 3rd ed. New York: McGraw-Hill Book Company, 1980. With permission.